SUMMARY

The objective of this project was to predict house prices using machine learning techniques. The dataset was cleaned by handling missing values, removing duplicates, and converting categorical variables into numerical form using one-hot encoding. Exploratory analysis revealed that area, bathrooms, stories, and air conditioning were among the strongest factors influencing house prices.

Two machine learning models were trained and evaluated: Linear Regression and Random Forest Regressor. The Linear Regression model achieved an R^2 score of approximately 0.65 and outperformed the Random Forest model on this dataset. This indicates that the linear model explained around 65% of the variation in house prices and provided reasonably accurate predictions.

Visualizations such as the price distribution histogram, correlation heatmap, and feature importance chart helped identify key trends and relationships in the data. Based on the findings, real estate businesses should focus on larger homes with desirable amenities such as multiple bathrooms and air conditioning, as these features significantly increase property value.